Economia das Organizações e dos Incentivos — MPE Insper

Exercise List: Real Book Exercises, Lecture 3

How to use this list. Every exercise below is a real exercise from one of the course books, reproduced verbatim: the source line gives author, book, chapter, exercise, the printed page and, in brackets, the page your PDF viewer shows for the file in `Livros/`, and whether the source gives the answer. Muñoz-Garcia & Toro-Gonzalez print a worked answer right after each exercise; MWG answers are in the MWG solutions manual in `Livros/`; Tadelis, Gibbons and Cabral give no answer in the book. When an exercise refers to a model, game or figure stated elsewhere in its book, that material is copied into the same block under "Referenced material", so every exercise can be solved where it is printed. There is one block per lecture of the real schedule. Each block has ten exercises in strictly rising difficulty, rated from 1 (apply a definition) to 5 (graduate level) next to each heading: three foundations, five exam level, two advanced. Try each exercise in the space provided before opening the book's answer.

Difficulty path. Lecture 3: from counting strategies (1) through Nature, imperfect recall and the zero-sum proof (MWG 9.B.4) to Gibbons 2.1 (5), a proof with general concave income and utility functions and continuous actions.

Block	#	Source	p. [PDF]	Level	What it trains	Diff.	Answer
L3	1 (21)	Tadelis 7.1‡	147 [166]	Found.	Counting pure strategies over information sets	1	none
	2 (22)	Muñoz-Garcia Ch. 4 Ex. 3	160 [172]	Found.	Sequential prisoner's dilemma	1	book
	3 (23)	Tadelis 7.2†	147 [166]	Found.	Story to tree; strategies; an "unappealing" Nash equilibrium	2	none
	4 (24)	Tadelis 8.5†	170 [189]	Exam	A simultaneous game behind an adoption choice: SPE	3	none
	5 (25)	Tadelis 8.10	172 [191]	Exam	Subgames with a simultaneous stage	3	none
	6 (26)	Tadelis 7.4‡	148 [167]	Exam	Four-move centipede: strategies, Nash equilibria, outcomes	3	none
	7 (27)	Tadelis 7.9‡	149–150 [168–169]	Exam	A move of Nature and an information set	4	none
	8 (28)	Tadelis 7.10†	150 [169]	Exam	Mixed vs. behavioral strategies; perfect recall	4	none
	9 (29)	MWG 9.B.4‡	302 [162]	Adv.	Proof: unique SPNE payoffs in zero-sum games of perfect information	4	manual
	10 (30)	Gibbons 2.1	130 [137]	Adv.	Rotten Kid theorem (continuous actions)	5	none

#: item number in this lecture's list, then its number in the combined list (`lista-reais-lectures-01-05.pdf`) in brackets. Items not in the list for Lectures 1–4 (`lista-reais-lectures-01-04.tex`) are marked: † added in the 30-item version, ‡ added in the 50-item version. Every other item is reused from that list with its statement unchanged, some in a different block or position. "Diff.": difficulty from 1 (apply a definition) to 5 (graduate level). Pages: printed page, then the PDF viewer page in brackets (offsets in `Livros/book-registry.json`; MWG is a two-pages-per-sheet scan). "Answer": *book* = worked answer printed after the exercise; *manual* = MWG solutions manual in `Livros/`; *none* = no answer in the book. Books: Tadelis (2013), *Game Theory: An Introduction*; Gibbons (1992), *Game Theory for Applied Economists*; Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, 2nd ed.; Mas-Colell, Whinston & Green (1995), *Microeconomic Theory* (MWG); Cabral (2017), *Introduction to Industrial Organization*, 2nd ed.

Block 3 — Dynamic games: extensive form, backward induction, SPE *(Lecture 3)*

Exercise 1. (item 21 of the combined list) *Strategies* [Foundations; difficulty 1/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 7 "Preliminaries", Exercise 7.1 "Strategies", p. 147 [PDF 166]. No answer in the book. Reproduced verbatim.

7.1 Strategies: Imagine an extensive-form game in which player i has K information sets.

a. If the player has an identical number of m possible actions in each information set, how many pure strategies does he have?

b. If the player has m_k actions in information set $k \in \{1, 2, \ldots, K\}$, how many pure strategies does the player have?

Resolution:

Exercise 2. (item 22 of the combined list) *Sequential Version of the Prisoner's Dilemma Game* [Foundations; difficulty 1/5]

Source. Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, 2nd ed., Ch. 4 "Sequential-Move Games with Complete Information", Exercise 3 "Sequential Version of the Prisoner's Dilemma Game"[A], p. 160 [PDF 172]. Worked answer in the book, right after the statement. Reproduced verbatim (statement only); Fig. 4.9 cropped from the page.

Consider the game tree in Fig. 4.9, representing the sequential-move version of the Prisoner's dilemma game, where player 1 acts first, and observing its move player 2 responds.

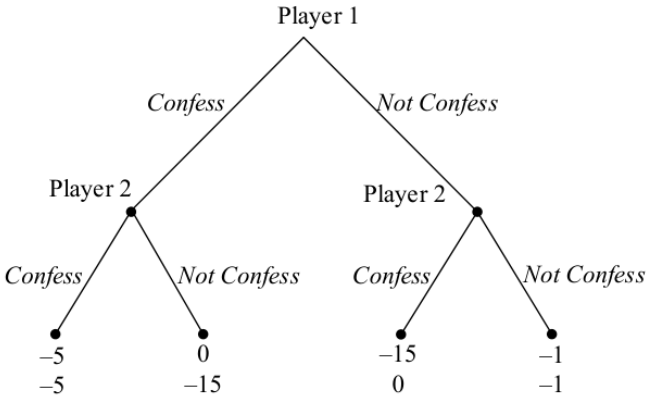


Fig. 4.9 Prisoner's dilemma sequential-move game

(a) Find the best responses of the second mover, for each of the first mover's actions.

(b) Find the first mover's equilibrium action and describe the subgame perfect Nash equilibrium of the game.

(c) Does the equilibrium behavior in the sequential-move game differ from its simultaneous-move version?

Referenced material. Source. "its simultaneous-move version": Muñoz-Garcia & Toro-Gonzalez (2019), Ch. 1, Exercise 5 "Prisoners' Dilemma Game", Fig. 1.8, p. 9 [PDF 22]. Matrix rebuilt from the page.

		Prisoner 2	
		Confess	Not confess
Prisoner 1	Confess	-5, -5	0, -15
	Not confess	-15, 0	-1, -1

Fig. 1.8 Normal-form of Prisoners' dilemma game

Resolution:

Exercise 3. (item 23 of the combined list) *Strategies and Equilibrium* [Foundations; difficulty 2/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 7 "Preliminaries", Exercise 7.2 "Strategies and Equilibrium", p. 147 [PDF 166]. No answer in the book. Reproduced verbatim.

7.2 Strategies and Equilibrium: Consider a two-player game in which player 1 can choose A or B. The game ends if he chooses A while it continues to player 2 if he chooses B. Player 2 can then choose C or D, with the game ending after C and continuing again with player 1 after D. Player 1 can then choose E or F, and the game ends after each of these choices.

a. Model this as an extensive-form game tree. Is it a game of perfect or imperfect information?

b. How many terminal nodes does the game have? How many information sets?

c. How many pure strategies does each player have?

d. Imagine that the payoffs following choice A by player 1 are $(2, 0)$, those following C by player 2 are $(3, 1)$, those following E by player 1 are $(0, 0)$, and those following F by player 1 are $(1, 2)$. What are the Nash equilibria of this game? Does one strike you as more "appealing" than the other? If so, explain why.

Resolution:

Exercise 4. (item 24 of the combined list) *Technology Adoption* [Exam level; difficulty 3/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 8 "Credibility and Sequential Rationality", Exercise 8.5 "Technology Adoption", p. 170 [PDF 189]. No answer in the book. Reproduced verbatim.

8.5 Technology Adoption: During the adoption of a new technology a CEO (player 1) can design a new task for a division manager. The new task can be either high level (H) or low level (L). The manager simultaneously chooses to invest in good training (G) or bad training (B). The payoffs from this interaction are given by the following matrix:

		Player 2	
		G	B
Player 1	H	5, 4	-5, 2
	L	2, -2	0, 0

a. Present the game in extensive form (a game tree) and solve for all the Nash equilibria and subgame-perfect equilibria.

b. Now assume that before the game is played the CEO can choose not to adopt this new technology, in which case the payoffs are (1, 1), or to adopt it, in which case the game is played. Present the *entire* game in extensive form. How many proper subgames does it have?

c. Solve for all the Nash equilibria and subgame-perfect equilibria of the game described in (b).

Resolution:

Exercise 5. (item 25 of the combined list) *Playing It Safe* [Exam level; difficulty 3/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 8 "Credibility and Sequential Rationality", Exercise 8.10 "Playing It Safe", p. 172 [PDF 191]. No answer in the book. Reproduced verbatim.

8.10 Playing It Safe: Consider the following dynamic game: Player 1 can choose to play it safe (denote this choice by S), in which case both he and player 2 get a payoff of 3 each, or he can risk playing a game with player 2 (denote this choice by R). If he chooses R then they play the following simultaneous-move game:

		Player 2	
		A	B
Player 1	C	8, 0	0, 2
	D	6, 6	2, 2

a. Draw a game tree that represents this game. How many proper subgames does it have?

b. Are there other game trees that would work? Explain briefly.

c. Construct the matrix representation of the normal form of this dynamic game.

d. Find all the Nash and subgame-perfect equilibria of the dynamic game.

Resolution:

Exercise 6. (item 26 of the combined list) *Centipedes* [Exam level; difficulty 3/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 7 "Preliminaries", Exercise 7.4 "Centipedes", p. 148 [PDF 167]. No answer in the book. Reproduced verbatim.

7.4 Centipedes: Imagine a two-player game that proceeds as follows. A pot of money is created with \$6 in it initially. Player 1 moves first, then player 2, then player 1 again, and finally player 2 again. At each player's turn to move, he has two possible actions: grab (G) or share (S). If he grabs he gets $\frac{2}{3}$ of the current pot of money, the other player gets $\frac{1}{3}$ of the pot, and the game ends. If he shares then the size of the current pot is multiplied by $\frac{3}{2}$ and the next player gets to move. At the last stage at which player 2 moves, if he chooses S then the pot is still multiplied by $\frac{3}{2}$, player 2 gets $\frac{1}{3}$ of the pot, and player 1 gets $\frac{2}{3}$ of the pot.

a. Model this as an extensive-form game tree. Is it a game of perfect or imperfect information?

b. How many terminal nodes does the game have? How many information sets?

c. How many pure strategies does each player have?

d. Find the Nash equilibria of this game. How many outcomes can be supported in equilibrium?

e. Now imagine that at the last stage at which player 2 moves, if he chooses to share then the pot is equally split among the players. Does your answer to part (d) change?

Resolution:

Exercise 7. (item 27 of the combined list) *The Dean's Dilemma* [Exam level; difficulty 4/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 7 "Preliminaries", Exercise 7.9 "The Dean's Dilemma", pp. 149–150 [PDF 168–169]. No answer in the book. Reproduced verbatim.

7.9 The Dean's Dilemma: A student stole the DVD player from the student lounge. The dean of students (player 1) suspects the student (player 2) and begins collecting evidence. However, evidence collection is a random process, and concrete evidence will be available to the dean only with probability $\frac{1}{2}$. The student knows that the evidence-gathering process is under way but does not know whether the dean has collected evidence or not. The game proceeds as follows: The dean realizes whether or not he has evidence and can then choose his action, whether to accuse the student (A) or bounce the case (B) and forget it. Once accused, the student has two options: he can either confess (C) or deny (D). Payoffs are realized as follows: If the dean bounces the case then both players get 0. If the dean accuses the student and the student confesses, the dean gains 2 and the student loses 2. If the dean accuses the student and the student denies, then payoffs depend on the evidence: If the dean has no evidence then he loses face, which gives him a loss of 4, while the student gains glory, which gives him a payoff of 4. If, however, the dean has evidence then he is triumphant and gains 4, while the student is put on probation and loses 4.

a. Draw the game tree that represents the extensive form of this game.

b. Write down the matrix that represents the normal form of the extensive form you drew in (a).

c. Solve for the Nash equilibria of the game.

Resolution:

Exercise 8. (item 28 of the combined list) *Perfect and Imperfect Recall* [Exam level; difficulty 4/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 7 "Preliminaries", Exercise 7.10 "Perfect and Imperfect Recall", p. 150 [PDF 169]. No answer in the book. Reproduced verbatim; Figure 7.14 cropped from the page.

7.10 Perfect and Imperfect Recall: Consider the game depicted in Figure 7.14.

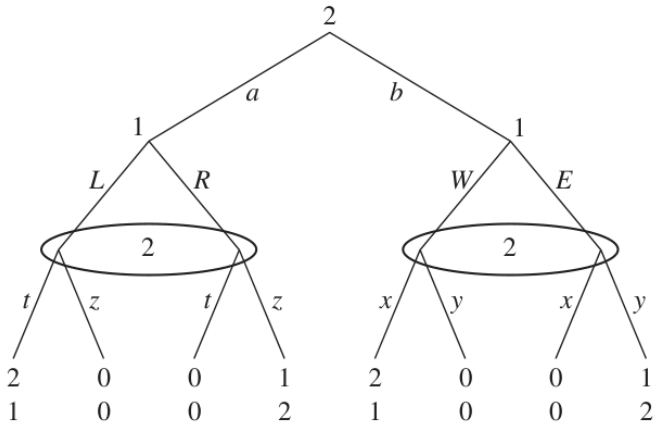


FIGURE 7.14 Exercise 7.10.

a. What are the pure-strategy sets for each player?

b. Show that for any behavioral strategy for player 1 there is a mixed strategy that leads to the same distribution over the terminal nodes regardless of the strategy chosen by player 2.

c. Show that for any behavioral strategy for player 2 there is a mixed strategy that leads to the same distribution over the terminal nodes regardless of the strategy chosen by player 1.

d. Now imagine that the game does not have perfect recall so that player 2's bottom two information sets are now one large information set. Can you find an example showing that the claim in (c) is no longer true?

Resolution:

Exercise 9. (item 29 of the combined list) *Zero-sum games of perfect information*

[Advanced; difficulty 4/5]

Source. Mas-Colell, Whinston & Green (1995), *Microeconomic Theory*, Ch. 9 "Dynamic Games", Exercises, Exercise 9.B.4^{B}, p. 302 [PDF 162, left half]. Answer in the MWG solutions manual (`Livros/`). Reproduced verbatim.

9.B.4^{B} Prove that in a finite *zero-sum* game of perfect information, there are unique subgame perfect Nash equilibrium payoffs.

Resolution:

Exercise 10. (item 30 of the combined list) *The Rotten Kid Theorem* [Advanced; difficulty 5/5]

Source. Gibbons (1992), *Game Theory for Applied Economists*, Ch. 2 "Dynamic Games of Complete Information", §2.6 Problems (Section 2.1), Problem 2.1, p. 130 [PDF 137]. No answer in the book. Reproduced verbatim.

2.1. Suppose a parent and child play the following game, first analyzed by Becker (1974). First, the child takes an action, A, that produces income for the child, $I_C(A)$, and income for the parent, $I_P(A)$. (Think of $I_C(A)$ as the child's income net of any costs of the action A.) Second, the parent observes the incomes I_C and I_P and then chooses a bequest, B, to leave to the child. The child's payoff is $U(I_C + B)$; the parent's is $V(I_P - B) + kU(I_C + B)$, where $k > 0$ reflects the parent's concern for the child's well-being. Assume that: the action is a nonnegative number, $A \geq 0$; the income functions $I_C(A)$ and $I_P(A)$ are strictly concave and are maximized at $A_C > 0$ and $A_P > 0$, respectively; the bequest B can be positive or negative; and the utility functions U and V are increasing and strictly concave. Prove the "Rotten Kid" Theorem: in the backwards-induction outcome, the child chooses the action that maximizes the family's aggregate income, $I_C(A) + I_P(A)$, even though only the parent's payoff exhibits altruism.

Resolution: